

V. SUMMARY

In this paper we have discussed, pedagogically, three increasingly more sensitive cases for solving the radial Dirac equations numerically:

1. A Lorentz scalar potential, V_s , which is linearly confining for quarks. The case of massless quarks is necessarily relativistic. As the mass of the quark increases, the wave function solutions become more and more non-relativistic.
2. A Lorentz vector potential, V_v , whose time-like component is a Coulomb potential. We first treated, numerically, the hydrogen atom, which is basically non-relativistic problem. We do, however, obtain the relativistic spin-orbit splitting between the $2p_{\frac{1}{2}}$ and $2p_{\frac{3}{2}}$ states (at the edge of our machine precision).
3. The combination of scalar and vector potentials, aiming at a relativistic quark model of mesons. Of particular interest is when we do (and do not) get a relativistic spin-orbit splitting.

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